

Predictors as Markets

One maker, makers in parallel, markets in series

Peter Cotton

Working draft v0.11 · August 29, 2026

Abstract

There are two algebraic reasons markets appear inside learning: convex duality turns optimization into trading, and graphical factorization turns inference into networks of local markets. We develop both. For one maker, non-convexity does not obstruct coherence: no-arbitrage is a chord condition, rational flow trades the biconjugate, and a proportional fee of at least the arbitrage depth (the worst per-unit excursion of chord slopes beyond the payoff hull) restores no-arbitrage, the fee being exactly a bid-ask spread by conjugation. For makers in parallel, combining fee-bearing makers is an infimal convolution solved by one monotone clearing-price root-find with sparse fills, and the aggregate supply curve is a consolidated limit order book. For markets in series we separate three things usually run together: a min-plus routing algebra, in which parallel merge multiplies factors and cheapest routing eliminates variables for arbitrary potentials; its market implementation, which we exhibit for quadratic makers on unrestricted real securities, where the alternation of model and market steps is exactly the Kalman filter and market messages give exact tree marginals; and probabilistic sum-product inference, which the routing algebra reproduces on log-quadratic families, where partial minimization equals marginalization up to a constant. A finite-state chain runs Viterbi in the routing algebra; we do not claim a coherent cost-function implementation of arbitrary finite-state kernels. A cost-based predictor admits a coherent market implementation if and only if its arbitrage depth is dominated by the permitted friction; friction extends the construction beyond convexity, and separation turns feasibility violations into portfolios that pay their discoverer. The resulting object is the incentive closure of the predictor: the same operator, with every information source paying to move the state.

1 Many predictors are already markets

A cost-function market maker posts a potential C over its inventory q of outstanding shares and charges $C(q + s) - C(q)$ for a fill of s shares; prices are gradients of C , and when C is convex the standard theory applies ([Hanson 2003](#); [Abernethy et al. 2013](#)). A run of published identities makes many learning procedures markets in this literal sense. Trading against such a maker is follow-the-regularized-leader with the cost as conjugate regularizer ([Chen and Vaughan 2010](#); [Abernethy et al. 2013](#)), with trader wealth as the learning rate ([Frongillo et al. 2012](#)). A market of Kelly bettors performs Bayesian model averaging with wealths as posterior weights ([Beygelzimer et al. 2012](#)), and equilibrium prices of utility-maximizing agents implement mixtures and products of experts ([Pennock and Wellman 1997](#); [Storkey 2011](#); [Storkey et al. 2012](#); [Barbu and Lay 2012](#)).

Two further identities are used throughout. Ordinary least squares is a market of data points: each observation a quadratic maker quoting its value with capital equal to its precision (inverse

variance), the estimate their merge, since merging makers is the infimal convolution of their costs and liquidity, the inverse of price impact, adds (Bhaskara et al. 2023; Barrieu and El Karoui 2005). And a proximal step is a trade against a fee-bearing maker: the proximal operator of $f|\cdot|$ is the soft-threshold, which is the optimal response to a proportional fee (Lemma 4 below), and the prox of any convex g is the response to a unit quadratic maker charging $\frac{1}{2}s^2 + g(s)$, since $\text{prox}_g(x) = \arg \max_s \{xs - \frac{1}{2}s^2 - g(s)\}$; the quadratic term is the maker's own curvature, and without it the trader's problem is a bare conjugate, unbounded for $|\mu| > f$ in the ℓ_1 case. Lemma 4 carries the base cost g_q for exactly this reason. The two canonical penalties are then the two market primitives: ridge is a zero-quoting participant with capital λ , lasso is a fee of λ , and the theorem that regularization is robustness to data perturbation (El Ghaoui and Lebrete 1997; Xu et al. 2009) acquires a market reading in which the adversary's budget is priced rather than assumed.

We ask the converse: which predictors are markets, and what does the market form demand? The answer has two independent parts, and the paper is organized around them. Convex duality turns optimization into trading: sections 2–4 treat one maker and sections 5–7 makers in parallel. Graphical factorization turns inference into networks of local markets: sections 8–11 treat markets in series, with the precise semiring statement in section 8. Section 12 states the characterization.

Two payoff regimes appear and must not be conflated. In the *bounded* regime a scalar security settles at $\varphi(\omega) = \omega \in [-1, 1]$, vector statements substituting the convex hull of payoff vectors; this is where Proposition 1 bites, where bounded worst-case loss is meaningful, and where sections 2–5 live. In the *unrestricted* regime the security settles at a real-valued quantity with payoff hull $\mathbb{R}$, so the chord condition is vacuous for finite positions and a quadratic maker $C(q) = mq + q^2/(2\lambda)$, whose chord slopes are unbounded, is admissible; this is the estimation regime of the Gaussian fusion, Kalman and tree results (§§6, 10–11) and of Proposition 13. Coherence statements transfer between the regimes only through the hull that defines them. Reference implementations and numerical theorem tests accompany the paper in the mechanisms repository.

2 Coherence is a chord condition

Proposition 1 (no-arbitrage without convexity). *For a path-independent maker (one whose charge depends only on the inventory endpoints) with cost C , no convexity assumed, there is no outcome-independent profitable fill from any state if and only if every chord slope $[C(q + s) - C(q)]/s$ lies in the convex hull of payoffs: for all q, s ,*

$$C(q + s) - C(q) \geq \min_{\omega} \varphi(\omega) s \quad (\text{scalar: } C \text{ is 1-Lipschitz}).$$

Proof. A fill s from state q has sure profit $\min_{\omega} \varphi(\omega)s - [C(q + s) - C(q)]$; positivity for some pair (q, s) is exactly the failure of the displayed inequality, and the inequality for all pairs says every chord slope is supported by the hull. ■

Round trips are refunds for any C , convex or not, because the charge telescopes over closed paths. When the chord condition fails the exploit is the accumulation of sure-profit net positions at states where quotes exit the hull.

Convexity enters the standard axiomatics through *information incorporation*, the marginal-cost monotonicity condition of Abernethy et al. (2013), not through no-arbitrage. The habit of

assuming convexity has bundled two different guarantees: monotone price response to flow, and absence of sure-loss opportunities. The axiom sets impose the first, convexity follows, and no-arbitrage is then derived with convexity in hand, so the possibility of a coherent maker with non-monotone quotes never arises. Proposition 1 separates the guarantees: dropping information incorporation while keeping the chord condition leaves a coherent non-convex maker, exhibited numerically in the companion repository. Arbitrage theory without convexity is developed in a different formalism by Lépinette and Tran (2017).

3 The market trades the biconjugate

Let $\hat{C}$ denote the lower convex envelope of a chord-coherent cost C (its biconjugate on the line (Rockafellar 1970)) and $g = C - \hat{C} \geq 0$ the gap.

Proposition 2 (contact and pass-through). *A myopic risk-neutral trader with believed mean μ in the hull, facing the maker at state q : (i) has every attained optimal fill landing on the contact set $\{C = \hat{C}\}$, an optimum existing whenever $x \mapsto C(x) - \mu x$ attains its infimum (for instance when it is coercive); and (ii) earns maximal expected profit*

$$\Pi_C(q, \mu) = \Pi_{\hat{C}}(q, \mu) + g(q)$$

as an identity of extended-real suprema, the envelope profit plus the gap at the starting state.

Proof. $\sup_s \mu s - [C(q + s) - C(q)] = \sup_x [\mu x - C(x)] - \mu q + C(q)$. Since $C \geq \hat{C}$ with equality on the contact set, and a maximizer of an affine function minus C is a point where an affine minorant touches C , hence touches $\hat{C}$, the supremum equals $\sup_x [\mu x - \hat{C}(x)]$, attained on the contact set when attained at all. Adding and subtracting $\hat{C}(q)$ gives (ii). ■

Attainment is a genuine hypothesis, not a formality. The chord-coherent cost $C(x) = \arctan x$ has lower convex envelope the constant $-\pi/2$ and hence empty contact set; at $\mu = 0$ the supremum $\pi/2$ is approached only as $x \rightarrow -\infty$. Where the relevant contact sits at infinity, the language below about flow landing on contact points and off-contact states being transient describes a limit that no fill realizes.

The maker behaves observationally like its convex envelope. Concave stretches are unquotable intermediate states, holes in the maker's book of standing quotes that rational flow jumps across. Whoever lands inside one (noise) overpays the gap at the landing state, and by (ii) the next rational trader recoups it, so the maker is a conduit keeping envelope differences over any rational-to-rational span, and off-contact states are transient. The trade-set analogue for constant-function market makers (CFMMs, the decentralized-exchange design) is the canonical concave trading function: an arbitrary invariant is behaviorally equivalent to a concave one (Angeris et al. 2024), with the limits of concavification mapped by Frongillo et al. (2024).

Read as beliefs, a hole is a multimodal quote: the maker takes prices on either side of the gap and refuses every price inside, jumping discontinuously as flow accumulates. This is the book-side shape of the bimodal implied densities documented around binary events (Melick and Thomas 1997; Clark and Amen 2017).

4 Frictions price the remaining failure

The failure mode surviving §2 is chord slopes exiting the hull.

Proposition 3 (fees buy bounded incoherence). *If chord slopes exit the hull by at most ε per unit, i.e. $C(q + s) - C(q) \geq \min_{\omega} \varphi(\omega)s - \varepsilon|s|$, then a proportional fee $f \geq \varepsilon$ restores no-arbitrage.*

Proof. Sure profit with the fee is $\min_{\omega} \varphi(\omega)s - \Delta C - f|s| \leq (\varepsilon - f)|s| \leq 0$. ■

This is the mechanism-level counterpart of a theorem of mathematical finance: under regularity conditions such as sticky paths or full support, arbitrarily small proportional transaction costs restore no-arbitrage for a large class of processes that are arbitrageable frictionlessly, with a consistent price system inside the spread as certificate (Guasoni 2006; Guasoni et al. 2010). In prediction markets the precedent is fees sized to expected arbitrage profit restoring bounded loss under privacy noise (Cummings et al. 2016; Frongillo and Waggoner 2018). Call the smallest ε for which the hypothesis of Proposition 3 holds the cost's *arbitrage depth*: the worst per-unit excursion of its chord slopes beyond the payoff hull, and so the minimum viable half-spread (the quoted spread being $2f$, since bid and ask sit at $m \mp f$). A venue quoting a wide spread to cover a deeply incoherent cost is arbitrage-free but uninformative in proportion: the market prices the model's incoherence as uncertainty.

5 A linear fee is a bid-ask spread

Why a fee at all? A cost-function maker's charge telescopes, so round trips are free: the maker earns nothing on uninformed flow and cannot recover from volume its adverse-selection losses, the losses to better-informed traders. That no state-dependent cost can charge a round trip, and that a path-dependent volume charge repairs it, is due to Othman and Sandholm (2012); the linear charge $f|s|$ is their volume levy in its simplest convex form, chosen because it conjugates in closed form. Write $g_q(s) = C(q + s) - C(q)$ for a convex C and $m = C'(q)$ for the marginal price.

Lemma 4 (fee–spread duality). *Let $T_f(x) = \text{sign}(x) \max(|x| - f, 0)$ denote the soft-threshold. Then the fee-bearing cost $\tilde{C}_q = g_q + f|\cdot|$ has conjugate*

$$\tilde{C}_q^*(p) = g_q^*(m + T_f(p - m)),$$

zero if and only if $|p - m| \leq f$.

Proof. The conjugate of a sum of closed proper convex functions with overlapping relative interiors of domains is the infimal convolution of the conjugates (Rockafellar 1970, Thm. 16.4), and the conjugate of $f|\cdot|$ is the indicator of $[-f, f]$. Hence $\tilde{C}_q^*(p) = \min_{|u| \leq f} g_q^*(p - u)$, the minimum of the convex function g_q^* over $[p - f, p + f]$. Since $g_q^* \geq 0$ with equality exactly at m , the minimum is attained at the projection of m onto the interval, which is $m + T_f(p - m)$. ■

The maker quotes ask = $m + f$ and bid = $m - f$ and trades nothing in between: a proportional fee is a bid-ask spread, the same duality by which a proportional transaction cost confines the pricing functional to the bid-ask band (Jouini and Kallal 1995). Without convexity the exact description needs no conjugation at all:

Lemma 5 (the no-trade interval, exactly). *For any cost C (no convexity assumed), state q , and fee $f \geq 0$, write $d_q(s) = [C(q + s) - C(q)]/s$ for the chord slope. A belief μ admits no profitable trade if and only if*

$$\sup_{s < 0} d_q(s) - f \leq \mu \leq \inf_{s > 0} d_q(s) + f,$$

so the set of no-trade beliefs is exactly this interval intersected with the payoff hull. For differentiable convex C both bounds equal $C'(q)$ and the interval is Lemma 4's band $[m - f, m + f]$. Write $\Delta_q = \sup_{s < 0} d_q - \inf_{s > 0} d_q$. The untruncated interval is non-empty if and only if $2f \geq \Delta_q$; if in addition C is chord-coherent, both bounds lie in the hull, so their midpoint does and the intersection with the hull is non-empty under the same condition. If C is chord-coherent and Proposition 2's supremum is attained, then off the contact set $\Delta_q > 0$.

Proof. No profitable trade means $\mu s \leq C(q + s) - C(q) + f|s|$ for all s . Dividing by $s > 0$ gives $\mu \leq d_q(s) + f$; dividing by $s < 0$ reverses the inequality to $\mu \geq d_q(s) - f$; together these are the displayed interval, whose non-emptiness is $\sup_{s < 0} d_q - f \leq \inf_{s > 0} d_q + f$. Under chord coherence both one-sided bounds are limits of chord slopes and so lie in the hull, which is convex, so the midpoint of a non-empty interval lies in the hull. For the last claim, off the contact set Proposition 2 gives every belief a strictly positive frictionless profit, so the $f = 0$ interval is empty. ■

The hull intersection is not decoration. Without chord coherence $C(q) = 100q$ has $\Delta_q = 0$, so the untruncated interval $\{100\}$ is non-empty for every $f \geq 0$, yet no admissible belief in $[-1, 1]$ declines to trade: the maker is arbitrated from every state.

So friction does not merely widen an existing spread: a spread with $2f \geq \Delta_q$ fills the quote hole that non-convexity created, and a state inside the hole becomes tenable. Non-convexity creates holes, frictionless rational flow jumps them (§3), and a large enough spread stabilizes points inside them. For $C(q) = a \sin q$ at $q = 0$, a state sitting a above its flat envelope, the belief $\mu = 0$ admits no profitable trade exactly when $f \geq a$, though every belief profits there frictionlessly.

6 Makers in parallel

Let makers $i = 1..n$ hold inventories q_i with costs C_i that are differentiable and strictly convex, with C'_i onto the price range of interest so that $(C'_i)^{-1}$ is defined there; fees f_i of their choosing; and write $m_i = C'_i(q_i)$ for maker i 's marginal price and $\text{ask}_i = m_i + f_i$, $\text{bid}_i = m_i - f_i$ for its quotes, as in Lemma 4. Without differentiability and strict convexity the supply curves below are set-valued and the statements hold with $(C'_i)^{-1}$ read as a subgradient correspondence.

One normalization is needed before the merged object is a maker. The infimal convolution of the incremental costs need not vanish at zero: for two zero-fee quadratic makers $g_i(s) = m_i s + s^2/(2\lambda_i)$,

$$(g_1 \square g_2)(0) = -\frac{(m_1 - m_2)^2}{2(1/\lambda_1 + 1/\lambda_2)} < 0$$

whenever the makers' quotes differ, because zero net external demand still admits a profitable internal cross-trade. We therefore assume the component inventories have been cleared against each other and the effective cost normalized to vanish at the resulting origin; otherwise evaluating at $\Delta = 0$ repeatedly appears to pay the same internal arbitrage again and again.

Lemma 6 (combination). Let $\tilde{C} = \tilde{C}_1 \square \dots \square \tilde{C}_n$ and define each maker's supply

$$s_i(p) = \begin{cases} (C'_i)^{-1}(p - f_i) - q_i, & p \geq \text{ask}_i, \\ 0, & \text{bid}_i < p < \text{ask}_i, \\ (C'_i)^{-1}(p + f_i) - q_i, & p \leq \text{bid}_i, \end{cases}$$

each non-decreasing in p . Fix a demand Δ for which a clearing price p^* with $\sum_i s_i(p^*) = \Delta$ exists. Then: (i) $\tilde{C}^* = \sum_i \tilde{C}_i^*$, a sum of soft-thresholded profit functions; (ii) the split $s_i = s_i(p^*)$ attains $\tilde{C}(\Delta)$, and any optimal split satisfies $C'_i(q_i + s_i) + f_i \text{sign}(s_i) = p^*$ for $s_i \neq 0$ and $|p^* - m_i| \leq f_i$ for $s_i = 0$; (iii) the split is sparse: every maker whose quote band strictly contains p^* trades exactly zero.

Proof. (i) is the conjugate-sum identity applied to the convolution (Rockafellar 1970). For (ii), the split is feasible by choice of p^* , and $p^* \in \partial \tilde{C}_i(s_i(p^*))$ for every i : when $s_i(p^*) \neq 0$ the subgradient is $C'_i(q_i + s_i) + f_i \text{sign}(s_i) = p^*$, and when $s_i(p^*) = 0$ it is the interval $[m_i - f_i, m_i + f_i] \ni p^*$. A common multiplier certifying every coordinate is exactly the optimality condition for $\min\{\sum_i \tilde{C}_i(s_i) : \sum_i s_i = \Delta\}$. (iii) restates the zero branch. ■

The computation is a scalar monotone root-find whatever n is, and the fee costs nothing beyond a horizontal shift of each supply curve. The $|s|$ terms are an ℓ_1 penalty, so sparsity arrives for the same reason it does in the lasso, and for the same reason proportional transaction costs produce no-trade regions and sparse portfolios (Oliveras-Nadal and DeMiguel 2018). A small trade routes to whichever maker posts the best quote, and to that maker alone unless the best quote is tied; a growing trade pushes that maker's fee-adjusted marginal price through the next band and spills over, consuming makers in quote-price order, which coincides with fee order only when their marginal prices agree.

Corollary 7 (zero fees). With $f_i \equiv 0$ the convolution reduces to the fee-free merge: conjugate regularisers add, and for a perspective family $C_b(q) = b C_1(q/b)$ liquidity adds, $C_{b_1} \square C_{b_2} = C_{b_1+b_2}$ (Bhaskara et al. 2023). For quadratic makers the merge is Gaussian fusion: the merged quote is the precision-weighted mean of the makers' quotes, and the merged liquidity is the sum of their liquidities (precisions).

Corollary 8 (the order book). The aggregate supply $S(p) = \sum_i s_i(p)$ is non-decreasing, identically zero on $(\max_i \text{bid}_i, \min_i \text{ask}_i)$ when that interval is non-empty, flat wherever every maker's band covers p , and smooth and strictly increasing wherever some maker is active. Read as a market: best bid and ask are the tightest quotes, $S(p)$ is the cumulative quantity executable up to price p with local depth $S'(p)$ where it exists, and large orders walk the levels. The aggregate of linear-fee makers is a consolidated limit order book, and in producer-theory terms Lemma 6 is Marshall's horizontal summation of firm supply curves (Marshall 1890; Mas-Colell et al. 1995) with the reversibility of share production patched by the fee. The economics of the book assembled from competing liquidity suppliers is Glosten (1994), with convergence of strategic schedules in Biais et al. (2000).

Proposition 9 (a deep co-quoter is Moreau smoothing). Merging a maker with cost C with a quadratic co-quoter of cost $s^2/(2\lambda)$ yields the venue with cost the Moreau envelope $e_\lambda C(x) = \min_y C(y) + (x - y)^2/(2\lambda)$. Then: (i) coherence is preserved: if the chord slopes of C lie in $[a, b]$, so do those of $e_\lambda C$, for every λ ; (ii) at any point where two distinct minimizing branches coexist (a crossing), the downward jump in marginal price is exactly the separation of the competing minimizers divided by λ , at most D/λ when all competing minimizers lie in an interval of diameter D ; (iii) for the symmetric double well $C(y) = \alpha \min(|y - c|, |y + c|)$ with $\alpha < 1$, the jump at $x = 0$ equals 2α while $\lambda\alpha < c$ and $2c/\lambda$ thereafter, so it is constant in the depth until the co-quoter is deep enough to reach across the wells. The bound in (ii) measures crossing defects only: the envelope can be smoothly non-convex with a unique minimizer everywhere, as for $C(y) = a \sin y$ with $\lambda a < 1$, where $(e_\lambda C)''(x) = C''(y^*)/(1 + \lambda C''(y^*))$ is negative wherever C'' is.

Proof. The merge is infimal convolution, and inf-convolution with the quadratic is the Moreau envelope (Rockafellar 1970). For (i), no differentiability is needed: for $h > 0$, sub-

stituting $y = z + h$,

$$e_\lambda C(x + h) = \inf_z \left\{ C(z + h) + \frac{(x - z)^2}{2\lambda} \right\} \leq e_\lambda C(x) + b h,$$

since $C(z + h) - C(z) \leq bh$ for every z ; the reverse substitution gives the lower bound ah , so every chord slope of $e_\lambda C$ lies in $[a, b]$. For (ii), at a crossing x_0 with competing minimizers $y_1^* < y_2^*$ the branch slopes are $(x_0 - y_i^*)/\lambda$, so the drop is exactly $(y_2^* - y_1^*)/\lambda$. For the smooth case, differentiate the first-order condition $y^* = x - \lambda C'(y^*)$ to get $dy^*/dx = 1/(1 + \lambda C''(y^*))$ and hence the displayed curvature. For (iii), the well's competing minimizers sit at $\pm\lambda\alpha$ while $\lambda\alpha < c$, giving separation $2\lambda\alpha$ and jump 2α , and at $\pm c$ once the co-quoter reaches the wells, giving $2c/\lambda$. ■

Part (i) is the durable statement: adding depth reshapes a non-convex maker without ever creating arbitrage, at any λ , so what the deep co-quoter buys is expressiveness. Crossing kinks close at rate $1/\lambda$ only where the competing minimizers stay within a λ -independent diameter; the double well shows what happens otherwise, its jump holding at 2α until the depth exceeds c/α . Smooth concave stretches lie outside the bound's scope entirely. Here λ is the co-quoter's liquidity: its price impact is $1/\lambda$, so large λ means a deep book, not a strong pull.

The two repairs are the two market primitives again: friction (ℓ_1 , the fee) and participant depth (ℓ_2 , capital), lasso and ridge. The identification maps the penalty terms, not the statistical procedures; it is a dictionary of primitives, not an equivalence of estimators.

7 Self-set fees and adaptivity

Nothing requires the fees to be administered. Each maker may quote its own f_i , and routing then disciplines the posted menu: a maker quoting inside the prevailing spread becomes the best quote and takes the flow first, a maker quoting behind the best price receives nothing until the book in front of it is consumed, and a maker quoting too tight is picked off by informed flow.

The discipline is a property of Lemma 6's routing, not of its minimization. The infimal convolution minimizes over allocations s_i with the fees held fixed; it does not minimize over the fees, and nothing in it constitutes a game in which makers choose f_i . What the surviving fee would be, and whether it is the competitive adverse-selection charge of classical microstructure (Glosten and Milgrom 1985; Biais et al. 2000), requires the strategic quote game, which is not solved here.

Optimization already pays these frictions: a proximal step charges $\|\Delta\theta\|^2/(2\eta)$ per move, a trust region is an infinite fee outside a band, weight decay is a zero-quoting participant. In market language the stabilizers of non-convex training are the market's two repairs: friction prices chord excursions (Proposition 3) and depth fills holes (Proposition 9); non-convexity by itself is not exploitability (§2). Path-dependent optimizers correspond to makers whose quotes depend on flow history, the adaptive-liquidity territory where path independence is deliberately traded away: no maker combines path independence, translation invariance (complete-bundle prices summing to the sure payoff) and liquidity sensitivity (Othman et al. 2013), the adaptive class is axiomatized by Li and Wortman Vaughan (2013) with the homogeneous-risk-measure characterization in Othman and Sandholm (2011), no trade-history maker achieves every desideratum at once (Abernethy et al. 2014), and liquidity selection itself can be run as online learning (Nueve et al. 2026; Nueve and Waggoner 2025).

8 Two operators, one semiring

Sections 5–7 composed makers on one quantity. A model is a factorization,

$$P(\text{everything}) = \prod_{\text{nodes}} P(x_i \mid \text{parents}(x_i)),$$

and the remaining operator runs one market per factor, each pricing its conditional given what is upstream. The count of two is structural. Message-passing inference is an algorithm over a commutative semiring (Aji and McEliece 2000): one operation combines evidence about a variable (the product), one moves evidence between variables (the sum), and sum-product, max-product and min-sum are the one algorithm over different semirings.

The correspondence with the market algebra is exact once the objects are fixed, and the fixing matters: the two compositions live on the two sides of the Legendre transform. Assign each factor its potential, the negative log-density $\varphi_i = -\log P(x_i \mid \text{parents})$. For parallel composition, identify each maker's factor potential with its *conjugate*, $\varphi_i = C_i^*$: infimal convolution of costs in inventory space is addition of potentials in price space. For serial composition, the stage potentials are the *primal* leg costs of a route. The dictionary is

factor algebra	$\longleftrightarrow$	market algebra
density p_i	$\longleftrightarrow$	potential $\varphi_i = -\log p_i$
multiply factors	$\longleftrightarrow$	parallel merge: $\varphi_i = C_i^*$ add as costs inf-convolve
eliminate a variable	$\longleftrightarrow$	route through its market: inf over the shared leg

with the two sides related by the Legendre transform, which is the Laplace transform of the min-plus semiring (Litvinov 2005). The parallel law is a commuting square,

$$\begin{array}{ccc} C_1, C_2 & \xrightarrow{\square} & C_1 \square C_2 \\ \downarrow * & & \downarrow * \\ C_1^*, C_2^* & \xrightarrow{+} & C_1^* + C_2^* \end{array}$$

while serial composition stays primal, obeying the composition law $(\varphi_2 \circ \varphi_1)(x, z) = \inf_y [\varphi_1(x, y) + \varphi_2(y, z)]$, the min-plus product of kernels (two-argument potentials). The Legendre transform is not a computational trick here but the change of representation between the two composition laws.

Proposition 10 (the routing algebra). (i) *Parallel: merging makers multiplies factors, since $(C_1 \square C_2)^* = C_1^* + C_2^*$ and potentials add exactly when densities multiply.* (ii) *Serial: the cheapest route to a terminal position through a chain of stage kernels costs $\inf_{z_1, \dots, z_{n-1}} \sum_i \varphi_i(z_{i-1}, z_i)$, the min-plus product of the kernels.* (iii) *On log-quadratic (Gaussian) families, partial minimization of a potential equals its marginalization up to an additive constant independent of the retained variables, so the min-plus messages are the sum-product messages.*

Proof. (i) is the conjugate-sum identity together with $-\log(p_1 p_2) = \varphi_1 + \varphi_2$. (ii) is the definition of the cheapest route: the trader chooses intermediate exposures to minimize the sum of stage costs. For (iii), write a jointly quadratic potential $q(x, y)$ with positive definite y -block Q_{yy} ; completing the square, $-\log \int e^{-q(x, y)} dy = \min_y q(x, y) + \frac{1}{2} \log \det(Q_{yy}/2\pi)$, the Schur-complement identity, and the constant does not depend on x . ■

Three things must be kept apart here, and the rest of the paper keeps them apart. The *routing algebra* is Proposition 10: min-plus identities that hold for arbitrary potentials, (ii) being little more than the definition of a cheapest route, and needing no convexity, as the generalized distributive law does not (Aji and McEliece 2000). *Sum-product inference* is what (iii) recovers on the log-quadratic family. A *market implementation* is a third thing, and strictly more: it requires each kernel to be posted as a path-independent maker with named securities and settlements, chord slopes inside the payoff hull, and the property that self-interested trades perform the minimization rather than a solver performing it. Proposition 10 does not supply that, and this paper exhibits it only for quadratic makers on unrestricted real securities, in §§10–11.

The implementability condition itself is the one §2 identified: a kernel is tradeable as a path-independent maker when its cost’s chord slopes respect the payoff hull. For closed convex C this reads dually as the potential $\varphi = C^*$ having effective domain inside the hull, so that $C = \varphi^*$ has subgradients there; a non-convex coherent maker is not its own biconjugate, $C^{**} = \hat{C} \neq C$, and instead shares this conjugate representation with its envelope, which is exactly the observational equivalence of §3. Convexity is the separate, further property of monotone information incorporation, and it is what gives the clean dual representation, not coherence.

Corollary (the serial algebra runs Viterbi). *Take finite state spaces and stage potentials $\varphi_t(i, j) = -\log P(X_t = j \mid X_{t-1} = i) - \log P(y_t \mid X_t = j)$. The cheapest route of Proposition 10(ii) is $\min_{x_{1:T}} \sum_t \varphi_t(x_{t-1}, x_t)$: the Viterbi decoding of the hidden Markov model, exactly, with no Gaussian structure anywhere. The statement is about the serial algebra; a coherent cost-function implementation of arbitrary finite-state kernels is not claimed. The three worked examples of this paper now form a progression: least squares is parallel composition and is implemented by quadratic makers, Viterbi is serial min-plus composition in the routing algebra alone, and the Kalman filter of §10 is the Gaussian intersection, where the serial computation is implemented by makers and is also Bayesian.*

Principle (the Gaussian intersection). *Log-quadratic families are a family on which probabilistic inference and min-plus optimization coincide, and one closed under both compositions: optimizing traders compute min-plus, an inference engine integrates, and Proposition 10(iii) says the two agree here up to constants that cancel in every price. The Kalman and tree-marginal propositions below live on this intersection and are not evidence for a universal serial thesis. The coincidence is not peculiar to quadratics: for $q(x, y) = x^2 + (y - x)^4$, minimization and marginalization also differ by an x -independent constant, so “log-quadratic” is sufficient and not necessary. Characterizing the families on which the two semirings agree, presumably by closure under a rich enough class of products, affine maps and eliminations, is open.*

Away from the intersection, min-plus elimination returns the max-marginal $x \mapsto \inf_y \varphi(x, y)$, the profile potential, rather than the sum-product marginal; the discrepancy is the Laplace-approximation gap. Note the max-marginal is a function, not the MAP point, which is its argmin. Risk aversion is an exponential tilt and plausibly interpolates between the two semirings; §12 poses this as the open question of which inference algorithm a risk-averse market runs.

9 One market per factor

The locality that makes the factorization tradable is Hanson’s modularity: in a combinatorial logarithmic market scoring rule (LMSR) a bet on $A \mid B$ moves that conditional and provably nothing else, uniquely among market scoring rules (Hanson 2007, 2003). Hanson runs one joint

market over the product space; the substance of pricing it is probabilistic inference, which is why exact pricing is #P-hard (Chen, Fortnow, et al. 2008), why tournament markets price by Bayes-net inference (Chen, Goel, et al. 2008), why a deployed combinatorial market ran its price and asset updates on the junction-tree algorithm (Sun et al. 2012), and why approximate designs price over the marginal polytope (Dudík et al. 2012; Dudík et al. 2021). Securities structured by a Bayes-net factorization appear in Pennock and Wellman (2000), with the trades that preserve the structure characterized by Xia and Pennock (2011).

The serial architecture takes the factors as separate venues rather than one joint book. Market i opens on $P(x_i \mid \text{parents})$ when its conditioning information freezes, that is, when the upstream quantities it conditions on stop changing; it conditions on upstream quotes while upstream is live, and re-references when upstream settles. Settlements cascade through the directed acyclic graph like dataflow, and the schedule of markets is an unrolling of the graph. Observable nodes carry settled markets; a latent node either settles through its observable footprint, a market on a hidden state being a market on a functional of future observables scored through the model, or remains indicated but unsettled, priced by whoever warehouses the model risk.

10 The market step is the Kalman update

Take the linear-Gaussian chain $x_{t+1} = ax_t + \varepsilon_t$, $y_t = x_t + \eta_t$, and alternate two steps: a *model step*, propagating the current belief through the dynamics in mean-precision form, and a *market step*, merging the propagated belief with an observation maker quoting y_t whose capital is the observation precision.

Proposition 11 (the market step is the Kalman update). *The alternation is the Kalman filter (Kalman 1960): after each observation the market state equals the filtered mean and variance exactly.*

Proof. The model step is the standard prediction of mean and variance. Merging quadratic makers adds precisions and precision-weights means (the fusion form of Corollary 7), which is the information-form measurement update; the covariance-form update follows by the usual algebra. ■

The prediction step is computed and the correction step is traded, with capital setting the weight of the data. The nearest published object is the Bayesian market maker of Brahma et al. (2012), whose trade update is a scalar Gaussian measurement update with covariance inflation for jumps; we find no filtering reading, and no alternation with a dynamic model, in the literature.

11 Trees and loops

On a Gaussian chain with an observation at every node, the posterior at an interior node t is the merge of three sources: the *predictive* forward message $p(x_t \mid y_{1:t-1})$, which is the filter of §10 run up to but not including y_t ; the local observation $p(y_t \mid x_t)$; and the backward message $p(y_{t+1:T} \mid x_t)$, downstream observations pulled back through the dynamics, each pull-back a reparametrization and each combination a precision addition. Taking the filtered posterior through node t as the forward message would count y_t twice.

Proposition 12 (trades as messages, exact on trees). *All three messages are market operations, and their merge equals the conditioning of the joint at every node: Gaussian belief propagation (Pearl 1988) with trades as messages.*

Proof. In information form the joint of a Gaussian chain factorizes into node and edge potentials. Induct on the chain: the predictive message at t is the model step applied to the merge at $t - 1$, a precision-scaled reparametrization; merging it with the local observation’s potential adds precisions (Corollary 7), which is the information-form conditioning identity; the backward recursion is the same argument run on the reversed chain, its pull-backs the Schur complements of Proposition 10(iii). Since each merge is exactly the information-form update and the three potentials multiply to the conditional, the merge is the posterior. ■

The repository verifies the identity to ten digits. The contrast with the deployed combinatorial engines is architectural: there the junction tree is the pricing algorithm inside one joint market (Sun et al. 2012); here the messages pass between venues, and the graph’s edges are market boundaries.

On loopy graphs belief propagation double-counts and its fixed points drift. In a market the same inconsistency is self-punishing.

Proposition 13 (quote inconsistency is a sure profit). *Let x be a vector of unrestricted real random variables and let separate venues quote securities paying the products $x_i x_j$ for every pair, the diagonal quoted at one, at fixed linear prices executable in the size traded; assemble the quotes into a matrix P with unit diagonal. The quotes admit a joint distribution iff $P \succeq 0$ (a Gaussian with covariance P witnesses sufficiency). Otherwise let w be a unit eigenvector of a negative eigenvalue $\lambda_{\min}$ and buy w_i^2 units of security ii and $2w_i w_j$ units of each security ij , $i < j$. The bundle costs $w^\top P w = \lambda_{\min} < 0$ and pays $(w^\top x)^2 \geq 0$: a sure profit of $|\lambda_{\min}|$. On three coordinates the quoted pairs are exactly the edges of a triangle, so the loopy reading is literal.*

This is the second-moment coherence condition, prices of products must form a PSD matrix (d’Aspremont 2005), read as cycle consistency. Arbitrage-enforced coherence is an old idea. Coherence is no-arbitrage (Nau and McCardle 1991); Pennock and Wellman (1996) built arbitrageur agents enforcing the additivity identities of a Bayes-net economy, with equilibrium prices equal to the network’s probabilities and distributed bidding as distributed inference; the combinatorial literature removes the arbitrage algorithmically over the marginal polytope (Kroer et al. 2016); and Saguillo et al. (2025) measure Polymarket arbitrageurs extracting \$39.6M enforcing logical coherence across markets.

Two boundaries of the proposition mark the general picture. If only a subgraph of pairs is quoted, the realizability condition is not positive semidefiniteness of a filled-in matrix but PSD-completeness of the partial one (Grone et al. 1984), and the arbitrage bundle must be supported on quoted pairs. If the coordinates are constrained, say binary $x_i \in \{\pm 1\}$, the feasible quote set is the cut polytope, a strict subset of the PSD body (Deza and Laurent 1997), so quotes can be positive semidefinite yet jointly unrealizable. In every case the coherent set is the convex hull of attainable payoffs, and Proposition 14 below is the universal form.

The scope of Proposition 13 is in any case one class of consistency failure: locally quoted beliefs that cannot be embedded in any joint distribution, detected here at second moments. Not every loopy-propagation error takes this form, but the class it covers is, in a market, free money, and the flow that harvests it pushes the quotes back toward the cone. Whether the corrected fixed point is the true marginal, a surrogate of the kind loopy propagation converges to, or something the liquidity profile selects is open.

The PSD cone is not special. Coherent price sets are convex (they are convex hulls of payoff vectors, or their conic images), and separation turns every exterior point into a trade:

Proposition 14 (arbitrage is separation). *Let K be the closed convex hull of the attainable payoff vectors (or its image under a linear security map), so that $\inf_{z \in K} \langle y, z \rangle \leq \langle y, \varphi(\omega) \rangle$ for every outcome*

ω and portfolio y , and let the venue quote fixed linear prices x , executable in the size traded. If $x \notin K$, any separating functional y with $\langle y, x \rangle < \inf_{z \in K} \langle y, z \rangle$ is a portfolio whose price is strictly less than its realized payoff in every state: a sure profit. Proposition 13 is the instance $K = \{P \succeq 0\}$.

For a nonlinear cost-function maker the conclusion is local. There x is the current gradient and a trade of size δy costs $\delta \langle y, x \rangle + o(\delta)$, since the marginal price moves as the trade is filled, so a strict separating margin yields sure profit for all sufficiently small δ , at a rate rather than in the size of y ; market impact and fees then set how large a position the certificate supports.

The proof is the separating-hyperplane theorem read as a trade (Nau and McCardle 1991; d’Aspremont 2005); the payoff-hull hypothesis is what upgrades the separation certificate to an arbitrage, and for an abstract consistency set the construction yields the certificate only. In an ordinary numerical method an infeasibility is a residual to be driven down by the algorithm; in a market it is a payoff, and whoever finds it is paid to act as the separation oracle. Arbitrageurs are decentralized separation oracles, and the friction of §4 sets the tolerance below which infeasibility is allowed to persist.

12 Closing

The characterization is for the cost-based class. Call a predictor *cost-based* if it is specified by a path-independent potential C over a security inventory, and call friction of size f *permitted* if the mechanism may charge up to f per unit traded. A cost-based predictor admits a coherent market implementation if and only if its arbitrage depth is dominated by the permitted friction; chord coherence is the $f = 0$ special case, and convexity is the separate, further property of monotone information incorporation. Three words are kept apart throughout: *coherence* (chord slopes stay in the payoff hull), *convexity* (marginal prices respond monotonically to flow), and *expressiveness* (the quoting range, which non-convexity restricts to the contact set). Within the coherent class, non-convexity spends expressiveness and spends nothing else against rational flow.

The construction throughout is one operation: *marketization*. Given an operator T from inputs to outputs, a marketization is a mechanism whose clearing computes T ; whose local contributions compose, in parallel and in series, to compute composite operators; in which inconsistent contributions create exploitable trades (Proposition 14); in which friction bounds how much inconsistency can persist (Proposition 3); and in which every participant pays to perturb the computation. One estimator becomes one maker, combining evidence becomes parallel composition, composing conditional operators becomes serial composition. A statistical procedure specifies how information would be combined if supplied honestly; its marketization implements the same operator against self-interested sources. In this sense a market is the incentive closure of a predictor: the computation, plus the dual certificates of its constraints, plus payment to whoever holds one. Duality alone supplies the middle column of

computation	certificate	payment
primal solve	dual multiplier	none
verified computation	proof of violation	none or fixed
incentive closure	separating portfolio	the certificate’s value

and the closure adds the third: a marketized computation is self-policing in proportion to the value of its own errors, with friction the exchange rate between tolerated error and the cost of participation.

The definition has neighbors in four literatures, and each holds one column. Markets that compute are old: a combinatorial auction cleared airport slots by integer programming (Rassenti et al. 1982), electricity markets pay the dual variables of the dispatch program as nodal prices (Schweppe et al. 1988), and market-oriented programming solved distributed allocation by computing competitive equilibria of artificial economies (Wellman 1993). Algorithmic mechanism design asks when optimization is compatible with incentives (Nisan and Ronen 2001), and the classification of what can be computed incentive-compatibly was posed as a program by Feigenbaum and Shenker (2002). The middle column carries a lower bound: any communication protocol realizing an efficient allocation must reveal supporting prices (Nisan and Segal 2006), so certificates are not optional. The third column exists in practice without the duality: fraud-proof systems pay whoever exhibits an invalid state transition (Teutsch and Reitwießner 2019; Kalodner et al. 2018), for discrete transitions rather than convex programs. And the friction bound has an incentive-free ancestor: the auction algorithm’s ε -complementary slackness permits suboptimality at most $n\varepsilon$, vanishing for $\varepsilon < 1/n$ on integer data (Bertsekas 1992); Proposition 3 is the same shape of bound with the ε charged to traders. The connective tissue, the closure operation itself, serial composition of mechanisms, and the identification of violation certificates with separating portfolios, appears unoccupied.

Nothing in the definition mentions prediction. The serial law is the min-plus kernel composition of dynamic programming, control, and shortest paths; the parallel law is the additive combination of local potentials; and the closure of the whole construction is a boundary statement:

Proposition 15 (the effective boundary maker). *Let a network of makers have boundary variables b , internal variables h , and local costs C_e , each closed, proper and jointly convex, with $\sum_e C_e(b, \cdot)$ coercive in h for each b in the boundary domain, and define $C_{\text{eff}}(b) = \inf_h \sum_e C_e(b, h)$. Then C_{eff} is convex, proper and finite there, the infimum is attained, and the elimination may be performed variable by variable in any order. For quadratic costs with positive definite internal block each single-variable elimination is a Schur complement, and on a chain the computation is the dynamic program of Proposition 10(ii). If in addition the chord slopes of C_{eff} lie in the boundary payoff hull, the network is one coherent effective maker at its boundary.*

Proof. Partial minimization of a jointly convex function is convex, coercivity gives attainment and rules out the value $-\infty$, and iterated infima may be taken in any order (Rockafellar 1970); the quadratic case is the completion of the square in the proof of Proposition 10(iii), which needs the internal block invertible. ■

Both hypotheses earn their place. A single local term linear and decreasing in an unconstrained internal variable drives $C_{\text{eff}} \equiv -\infty$; and convexity of C_{eff} says nothing about coherence against a bounded boundary hull, which is why the last clause is separate.

This is the operation that reduces resistor networks to terminal impedances, eliminates latent Gaussian variables, and takes Schur complements: internal competition disappears into an external price law, as internal nodes disappear into a Dirichlet-to-Neumann map. One caution attaches to “any order”: the effective boundary market is elimination-order invariant, but the computational cost of producing it is not, and the blowup of intermediate scopes is the treewidth phenomenon of variable elimination. Prediction is the application in which beliefs and prices share units. We leave the general theory, including the categorical formulation in which markets are min-plus kernels composed by shared inventory, outside this paper’s scope.

One principle deserves separation from the open problems it generates, because it qualifies everything above.

Principle (accessible arbitrage). *Market coherence is relative not only to friction but to the cost of discovering violations. Write K for the exactly coherent quotes (Proposition 1) and K_f for those admitting no sure profit net of a fee f (Proposition 3). Let $\mathcal{A}$ be a class of admissible certificate-search procedures, and for a quoted configuration x let $V_f(A(x); x)$ be the maximum realizable* profit from the portfolio a procedure A finds, computed against the venue’s actual books and net of trading friction, under the position or capital constraint the arbitrageur faces, and let $C_A(x)$ be the search cost. The accessible-coherence set is**

$$K_{\mathcal{A},f} = \left\{ x : \sup_{A \in \mathcal{A}} \mathbb{E} [V_f(A(x); x) - C_A(x)] \leq 0 \right\}.$$

With nonnegative search costs, $K \subseteq K_f \subseteq K_{\mathcal{A},f}$: a violation persists whenever every available separation procedure costs at least as much, net of friction, as the certificate it finds is worth.

Both features of V_f are load-bearing. The supremum ranges over procedures, not certificates: two procedures can find the same portfolio at very different costs, so coherence is relative to a class of arbitrageurs. And the profit must be the realizable one. A separating functional may be scaled by any positive constant, so against fixed linear prices its nominal value is unbounded and no finite search cost could ever leave a violation standing; it is the nonlinear book, the finite executable depth, and the position constraint that make $K_f \subsetneq K_{\mathcal{A},f}$ possible at all. Equivalently one may normalize certificates to $\|y\| \leq 1$ and price them against available depth. Instantiating $\mathcal{A}$ gives polynomial-time coherence, bounded-budget coherence, and latency-constrained coherence, and a venue can be arbitrage-free to ordinary participants while arbitrageable to a better-equipped class. Three nested notions of coherence result: exact (no sure-profit chord, Proposition 1), frictional (violations below the spread survive, Proposition 3), and accessible (violations too expensive to discover survive). Proposition 14 addresses the existence of arbitrage; whether a procedure in $\mathcal{A}$ can profitably prove $x \notin K$ is its accessibility, a different property. The self-policing of the marketization table is qualified accordingly: effective self-correction is the error’s value net of discovery cost and friction.

Open problems, in rough order of tractability:

Which inference algorithm does a risk-averse market run? Proposition 10 says risk-neutral routing computes min-plus exactly and sum-product only on the Gaussian intersection. Risk aversion is an exponential tilt, suggesting it interpolates between the two semirings; whether a risk-averse trading population prices the marginal, the mode, or a temperature in between would complete the correspondence.

A serial market mechanism in general. Proposition 10(ii) is routing algebra and §§10–11 implement it only for quadratic makers on unrestricted real securities. Give, for a useful class of kernels beyond the quadratic, the securities, settlements and inventory variables of each stage market, verify chord coherence stage by stage, and show that self-interested trading performs the elimination. The interface between the two compositions is part of the problem: parallel merge acts on conjugates and serial routing on primal leg costs, so a factor graph alternating the two needs an explicit change of representation at each junction.

Market representations of general prediction maps. The characterization above is confined to cost-based predictors. Define what it means for an arbitrary prediction map T , from data sets to forecasts, to possess a market representation, and give conditions on T equivalent to existence. A first conjecture: T is representable when it factors through a network of cost kernels whose boundary reduction is T , with the chord bound applying kernel by kernel; path independence and the chord condition would then be the image of the general condition in the cost-based case.

Accessible coherence, characterized. Compute $K_{\mathcal{A},f}$ for concrete classes: polynomial-time arbitrageurs against cones whose separation is hard, budgeted arbitrageurs against fee-bearing venues, latency-constrained arbitrageurs against cascading settlements. And characterize the mechanisms in which the stream of certificate payments funds the ongoing computation.

The equilibrium fee. Section 7 argues discipline, not equilibrium. In a flow model where most coefficients are zero and informed trades carry signal plus noise, does the fee that survives undercutting reproduce the universal threshold $\sigma (2 \log p)^{1/2}$ of the shrinkage literature? If so, the regularization constant that statistics tunes by cross-validation equals the adverse-selection cost of the data source.

Loopy fixed points. Under a concrete flow model, does arbitrage-corrected quoting converge, and to what?

Arbitrage depth as a capacity measure. The arbitrage depth of a trained model’s loss landscape is computable; whether it tracks quantities learning theory names (sharpness, mode connectivity) is open.

Effective friction. Proposition 15 eliminates frictionless networks. With per-maker fees, what are the effective fees and liquidities of the reduced boundary maker in terms of the network’s, and which fee-bearing classes are closed under elimination?

Books around binary events. The multimodal-quote reading of book holes (§3) is testable against limit-order data around court rulings and FDA decisions, where bimodal implied densities are documented (Clark and Amen 2017).

References

- Abernethy, Jacob, Yiling Chen, and Jennifer Wortman Vaughan. 2013. “Efficient Market Making via Convex Optimization, and a Connection to Online Learning.” *ACM Transactions on Economics and Computation* 1 (2): 12:1–39. <https://doi.org/10.1145/2465769.2465777>.
- Abernethy, Jacob, Rafael Frongillo, Xiaolong Li, and Jennifer Wortman Vaughan. 2014. “A General Volume-Parameterized Market Making Framework.” *Proceedings of the 15th ACM Conference on Economics and Computation (EC)*, 413–30. <https://doi.org/10.1145/2600057.2602900>.
- Aji, Srinivas M., and Robert J. McEliece. 2000. “The Generalized Distributive Law.” *IEEE Transactions on Information Theory* 46 (2): 325–43. <https://doi.org/10.1109/18.825794>.
- Angeris, Guillermo, Tarun Chitra, Theo Diamandis, Alex Evans, and Kshitij Kulkarni. 2024. “The Geometry of Constant Function Market Makers.” *Proceedings of the 25th ACM Conference on Economics and Computation (EC)*, 732. <https://doi.org/10.1145/3670865.3673500>.
- Barbu, Adrian, and Nathan Lay. 2012. “An Introduction to Artificial Prediction Markets for Classification.” *Journal of Machine Learning Research* 13: 2177–204. <https://arxiv.org/abs/1102.1465>.
- Barrieu, Pauline, and Nicole El Karoui. 2005. “Inf-Convolution of Risk Measures and Optimal Risk Transfer.” *Finance and Stochastics* 9 (2): 269–98. <https://doi.org/10.1007/s00780-005-0152-0>.

- Bertsekas, Dimitri P. 1992. "Auction Algorithms for Network Flow Problems: A Tutorial Introduction." *Computational Optimization and Applications* 1 (1): 7–66. <https://doi.org/10.1007/BF00247653>.
- Beygelzimer, Alina, John Langford, and David M. Pennock. 2012. "Learning Performance of Prediction Markets with Kelly Bettors." *Proceedings of the 11th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*. <https://arxiv.org/abs/1201.6655>.
- Bhaskara, Adithya, Rafael Frongillo, Elias Lindgren, and Maneesha Papireddygar. 2023. *A General Theory of Liquidity Provisioning for Prediction Markets*. arXiv:2311.08725. <https://arxiv.org/abs/2311.08725>.
- Biais, Bruno, David Martimort, and Jean-Charles Rochet. 2000. "Competing Mechanisms in a Common Value Environment." *Econometrica* 68 (4): 799–837. <https://doi.org/10.1111/1468-0262.00138>.
- Brahma, Aseem, Mithun Chakraborty, Sanmay Das, Allen Lavoie, and Malik Magdon-Ismael. 2012. "A Bayesian Market Maker." *Proceedings of the 13th ACM Conference on Electronic Commerce (EC)*, 215–32. <https://doi.org/10.1145/2229012.2229031>.
- Chen, Yiling, Lance Fortnow, Nicolas Lambert, David M. Pennock, and Jennifer Wortman. 2008. "Complexity of Combinatorial Market Makers." *Proceedings of the 9th ACM Conference on Electronic Commerce (EC)*, 190–99. <https://doi.org/10.1145/1386790.1386822>.
- Chen, Yiling, Sharad Goel, and David M. Pennock. 2008. "Pricing Combinatorial Markets for Tournaments." *Proceedings of the 40th ACM Symposium on Theory of Computing (STOC)*, 305–14. <https://doi.org/10.1145/1374376.1374421>.
- Chen, Yiling, and Jennifer Wortman Vaughan. 2010. "A New Understanding of Prediction Markets via No-Regret Learning." *Proceedings of the 11th ACM Conference on Electronic Commerce (EC)*, 189–98. <https://doi.org/10.1145/1807342.1807372>.
- Clark, Iain J., and Saeed Amen. 2017. "Implied Distributions from GBPUSD Risk-Reversals and Implication for Brexit Scenarios." *Risks* 5 (3): 35. <https://doi.org/10.3390/risks5030035>.
- Cummings, Rachel, David M. Pennock, and Jennifer Wortman Vaughan. 2016. "The Possibilities and Limitations of Private Prediction Markets." *Proceedings of the 17th ACM Conference on Economics and Computation (EC)*. <https://arxiv.org/abs/1602.07362>.
- d'Aspremont, Alexandre. 2005. *A Market Test for the Positivity of Arrow-Debreu Prices*. arXiv:cs/0510027. <https://arxiv.org/abs/cs/0510027>.
- Deza, Michel M., and Monique Laurent. 1997. *Geometry of Cuts and Metrics*. Vol. 15. Algorithms and Combinatorics. Springer. <https://doi.org/10.1007/978-3-642-04295-9>.
- Dudík, Miroslav, Xintong Wang, David M. Pennock, and David M. Rothschild. 2021. "Log-Time

- Prediction Markets for Interval Securities." *Proceedings of the 20th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 465–73. <https://doi.org/10.65109/bbzi6501>.
- Dudík, Miroslav, Sébastien Lahaie, and David M. Pennock. 2012. "A Tractable Combinatorial Market Maker Using Constraint Generation." *Proceedings of the 13th ACM Conference on Electronic Commerce (EC)*, 459–76. <https://doi.org/10.1145/2229012.2229047>.
- El Ghaoui, Laurent, and Hervé Lebrete. 1997. "Robust Solutions to Least-Squares Problems with Uncertain Data." *SIAM Journal on Matrix Analysis and Applications* 18 (4): 1035–64. <https://doi.org/10.1137/S0895479896298130>.
- Feigenbaum, Joan, and Scott Shenker. 2002. "Distributed Algorithmic Mechanism Design: Recent Results and Future Directions." *Proceedings of the 6th International Workshop on Discrete Algorithms and Methods for Mobile Computing and Communications (DIALM)*, 1–13. <https://www.cs.yale.edu/homes/jf/FS.pdf>.
- Frongillo, Rafael M., Nicolás Della Penna, and Mark D. Reid. 2012. "Interpreting Prediction Markets: A Stochastic Approach." *Advances in Neural Information Processing Systems* 25 (NeurIPS). <https://papers.nips.cc/paper/4529-interpreting-prediction-markets-a-stochastic-approach>.
- Frongillo, Rafael, Maneesha Papireddygar, and Bo Waggoner. 2024. "An Axiomatic Characterization of CFMMs and Equivalence to Prediction Markets." *15th Innovations in Theoretical Computer Science Conference (ITCS)*, LIPIcs, vol. 287: 51:1–21. <https://doi.org/10.4230/LIPIcs.ITCS.2024.51>.
- Frongillo, Rafael, and Bo Waggoner. 2018. "Bounded-Loss Private Prediction Markets." *Advances in Neural Information Processing Systems (NeurIPS)*. <https://arxiv.org/abs/1703.00899>.
- Glosten, Lawrence R. 1994. "Is the Electronic Open Limit Order Book Inevitable?" *The Journal of Finance* 49 (4): 1127–61. <https://doi.org/10.1111/j.1540-6261.1994.tb02450.x>.
- Glosten, Lawrence R., and Paul R. Milgrom. 1985. "Bid, Ask and Transaction Prices in a Specialist Market with Heterogeneously Informed Traders." *Journal of Financial Economics* 14 (1): 71–100.
- Grone, Robert, Charles R. Johnson, Eduardo M. Sá, and Henry Wolkowicz. 1984. "Positive Definite Completions of Partial Hermitian Matrices." *Linear Algebra and Its Applications* 58: 109–24. [https://doi.org/10.1016/0024-3795\(84\)90207-6](https://doi.org/10.1016/0024-3795(84)90207-6).
- Guasoni, Paolo. 2006. "No Arbitrage Under Transaction Costs, with Fractional Brownian Motion and Beyond." *Mathematical Finance* 16 (3): 569–82. <https://doi.org/10.1111/j.1467-9965.2006.00283.x>.
- Guasoni, Paolo, Miklós Rásonyi, and Walter Schachermayer. 2010. "The Fundamental Theorem

- of Asset Pricing for Continuous Processes Under Small Transaction Costs." *Annals of Finance* 6 (2): 157–91. <https://doi.org/10.1007/s10436-008-0110-x>.
- Hanson, Robin. 2003. "Combinatorial Information Market Design." *Information Systems Frontiers* 5 (1): 107–19. <https://doi.org/10.1023/A:1022058209073>.
- Hanson, Robin. 2007. "Logarithmic Market Scoring Rules for Modular Combinatorial Information Aggregation." *The Journal of Prediction Markets* 1 (1): 3–15. <https://doi.org/10.5750/jpm.v1i1.417>.
- Jouini, Elyès, and Hédi Kallal. 1995. "Martingales and Arbitrage in Securities Markets with Transaction Costs." *Journal of Economic Theory* 66 (1): 178–97. <https://doi.org/10.1006/jeth.1995.1037>.
- Kalman, Rudolf E. 1960. "A New Approach to Linear Filtering and Prediction Problems." *Journal of Basic Engineering* 82 (1): 35–45. <https://doi.org/10.1115/1.3662552>.
- Kalodner, Harry, Steven Goldfeder, Xiaoqi Chen, S. Matthew Weinberg, and Edward W. Felten. 2018. "Arbitrum: Scalable, Private Smart Contracts." *Proceedings of the 27th USENIX Security Symposium*, 1353–70. <https://www.usenix.org/conference/usenixsecurity18/presentation/kalodner>.
- Kroer, Christian, Miroslav Dudík, Sébastien Lahaie, and Sivaraman Balakrishnan. 2016. "Arbitrage-Free Combinatorial Market Making via Integer Programming." *Proceedings of the 17th ACM Conference on Economics and Computation (EC)*, 161–78. <https://arxiv.org/abs/1606.02825>.
- Lépinette, Emmanuel, and Tuan Quoc Tran. 2017. "Arbitrage Theory for Non Convex Financial Market Models." *Stochastic Processes and Their Applications* 127 (10): 3331–53. <https://www.sciencedirect.com/science/article/pii/S0304414915301216>.
- Li, Xiaolong, and Jennifer Wortman Vaughan. 2013. "An Axiomatic Characterization of Adaptive-Liquidity Market Makers." *Proceedings of the 14th ACM Conference on Electronic Commerce (EC)*, 657–74. <https://doi.org/10.1145/2482540.2482575>.
- Litvinov, Grigory L. 2005. *The Maslov Dequantization, Idempotent and Tropical Mathematics: A Brief Introduction*. arXiv:math/0507014. <https://arxiv.org/abs/math/0507014>.
- Marshall, Alfred. 1890. *Principles of Economics*. Macmillan.
- Mas-Colell, Andreu, Michael D. Whinston, and Jerry R. Green. 1995. *Microeconomic Theory*. Oxford University Press.
- Melick, William R., and Charles P. Thomas. 1997. "Recovering an Asset's Implied PDF from Option Prices: An Application to Crude Oil During the Gulf Crisis." *Journal of Financial and Quantitative Analysis* 32 (1): 91–115. <https://doi.org/10.2307/2331318>.

- Nau, Robert F., and Kevin F. McCardle. 1991. "Arbitrage, Rationality, and Equilibrium." *Theory and Decision* 31 (2–3): 199–240.
- Nisan, Noam, and Amir Ronen. 2001. "Algorithmic Mechanism Design." *Games and Economic Behavior* 35 (1–2): 166–96. <https://doi.org/10.1006/game.1999.0790>.
- Nisan, Noam, and Ilya Segal. 2006. "The Communication Requirements of Efficient Allocations and Supporting Prices." *Journal of Economic Theory* 129 (1): 192–224. <https://doi.org/10.1016/j.jet.2004.10.007>.
- Nueve, Enrique, Bao Nguyen, Rafael Frongillo, and Bo Waggoner. 2026. *Adaptive Liquidity in Prediction Markets via Online Learning*. arXiv:2605.09599. <https://arxiv.org/abs/2605.09599>.
- Nueve, Enrique, and Bo Waggoner. 2025. *Smooth Quadratic Prediction Markets*. arXiv:2505.02959. <https://arxiv.org/abs/2505.02959>.
- Olivares-Nadal, Alba V., and Victor DeMiguel. 2018. "A Robust Perspective on Transaction Costs in Portfolio Optimization." *Operations Research* 66 (3). <https://doi.org/10.1287/opre.2017.1699>.
- Othman, Abraham, David M. Pennock, Daniel M. Reeves, and Tuomas Sandholm. 2013. "A Practical Liquidity-Sensitive Automated Market Maker." *ACM Transactions on Economics and Computation* 1 (3). <https://doi.org/10.1145/2509413.2509414>.
- Othman, Abraham, and Tuomas Sandholm. 2011. "Liquidity-Sensitive Automated Market Makers via Homogeneous Risk Measures." *Internet and Network Economics (WINE 2011)*, Lecture notes in computer science, vol. 7090. https://doi.org/10.1007/978-3-642-25510-6_27.
- Othman, Abraham, and Tuomas Sandholm. 2012. "Profit-Charging Market Makers with Bounded Loss, Vanishing Bid/Ask Spreads, and Unlimited Market Depth." *Proceedings of the 13th ACM Conference on Electronic Commerce (EC)*, 790–807. <https://doi.org/10.1145/2229012.2229074>.
- Pearl, Judea. 1988. *Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference*. Morgan Kaufmann.
- Pennock, David M., and Michael P. Wellman. 1996. "Toward a Market Model for Bayesian Inference." *Proceedings of the 12th Conference on Uncertainty in Artificial Intelligence (UAI)*, 405–13. <https://arxiv.org/abs/1302.3593>.
- Pennock, David M., and Michael P. Wellman. 1997. "Representing Aggregate Belief Through the Competitive Equilibrium of a Securities Market." *Proceedings of the 13th Conference on Uncertainty in Artificial Intelligence (UAI)*, 392–400. <https://arxiv.org/abs/1302.1564>.
- Pennock, David M., and Michael P. Wellman. 2000. "Compact Securities Markets for Pareto Optimal Reallocation of Risk." *Proceedings of the 16th Conference on Uncertainty in Artificial*

- Intelligence (UAI)*, 481–88. <https://arxiv.org/abs/1301.3886>.
- Rassenti, Stephen J., Vernon L. Smith, and Robert L. Bulfin. 1982. “A Combinatorial Auction Mechanism for Airport Time Slot Allocation.” *The Bell Journal of Economics* 13 (2): 402–17. <https://doi.org/10.2307/3003463>.
- Rockafellar, R. Tyrrell. 1970. *Convex Analysis*. Princeton Mathematical Series 28. Princeton University Press. <https://doi.org/10.1515/9781400873173>.
- Saguillo, Oriol, Vahid Ghafouri, Lucianna Kiffer, and Guillermo Suarez-Tangil. 2025. “Unraveling the Probabilistic Forest: Arbitrage in Prediction Markets.” *Advances in Financial Technologies (AFT)*. <https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.AFT.2025.27>.
- Schweppe, Fred C., Michael C. Caramanis, Richard D. Tabors, and Roger E. Bohn. 1988. *Spot Pricing of Electricity*. Kluwer Academic Publishers.
- Storkey, Amos J. 2011. “Machine Learning Markets.” *Proceedings of the 14th International Conference on Artificial Intelligence and Statistics (AISTATS)*, PMLR, vol. 15: 716–24. <https://arxiv.org/abs/1106.4509>.
- Storkey, Amos J., Jono Millin, and Krzysztof Geras. 2012. “Isoelastic Agents and Wealth Updates in Machine Learning Markets.” *International Conference on Machine Learning (ICML)*. <https://arxiv.org/abs/1206.6443>.
- Sun, Wei, Robin Hanson, Kathryn B. Laskey, and Charles Twardy. 2012. “Probability and Asset Updating Using Bayesian Networks for Combinatorial Prediction Markets.” *Proceedings of the 28th Conference on Uncertainty in Artificial Intelligence (UAI)*, 815–24. <https://arxiv.org/abs/1210.4900>.
- Teutsch, Jason, and Christian Reitwießner. 2019. *A Scalable Verification Solution for Blockchains*. arXiv:1908.04756. <https://arxiv.org/abs/1908.04756>.
- Wellman, Michael P. 1993. “A Market-Oriented Programming Environment and Its Application to Distributed Multicommodity Flow Problems.” *Journal of Artificial Intelligence Research* 1: 1–23. <https://doi.org/10.1613/jair.2>.
- Xia, Lirong, and David M. Pennock. 2011. “Price Updating in Combinatorial Prediction Markets with Bayesian Networks.” *Proceedings of the 27th Conference on Uncertainty in Artificial Intelligence (UAI)*, 825–34. <https://arxiv.org/abs/1202.3756>.
- Xu, Huan, Constantine Caramanis, and Shie Mannor. 2009. “Robustness and Regularization of Support Vector Machines.” *Journal of Machine Learning Research* 10: 1485–510. <https://jmlr.org/papers/v10/xu09b.html>.